

As quantity grows when the requested corrections are large and spread over many independent sections. Crucially, it depends on both the dataset D and the frozen model M : changing the receiver changes its hidden states, its errors, and therefore the correction spectrum. After averaging seeds and removing exact data reuses, our comparison contains 22 distinct del-dataset conditions across seven task families. We evaluate each candidate both by how well it orders R^* within a model and by how well it predicts a task family left out of fitting. Because ten measures were screened on the same data, our reported permutation test also corrected for selecting the best candidate from that set.

Measure	Within-model ρ	Adjusted p	Held-out RMSE	Pooled ρ
Correction Fisher log-volume	0.764	0.00034	0.187	0.584
Correction Fisher effective rank	0.609	0.0112	0.181	0.814
Surprise-weighted hidden log-volume	0.614	0.0103	0.204	0.441
Base-model target code length	0.200	0.524	0.261	0.269
Model-only fit	—	—	0.264	—

Table 2: Screen of ten model-relative measures against likelihood-based $R^*(.90)$. RMSE leaves one task family out. ρ is the Spearman correlation, and p is the p-value of the permutation test.

The correction spectrum also follows the earlier SQL/XBRL contrast: its log-volume is nearly flat as SQL diversity increases but rises with XBRL diversity on both base models. It is not simply another name for learning gain; across the distinct conditions, correction log-volume and held-out gain have Spearman correlation -0.275 .

This is our strongest current predictor. It however uses a projected output-head gradient rather than the full LoRA Fisher, and the study contains only two 7B receivers. We therefore treat the correction spectrum as a promising model-relative hypothesis, not yet as an established scaling law. This is ongoing at the time of writing.